VoiceOps 2026

Technical submission report | 29 October 2026

Source repository: https://github.com/chukka-venugopalam/Voiceops_2026

Executive summary

VoiceOps is a voice-capable customer-support agent for the fictional store ZentroMart. The implementation combines section-based knowledge retrieval, deterministic business rules and calculations, conversation state, escalation/ticket workflows, optional LLM response wording, and Gnani speech input/output. The package includes runnable application entry points, tests, evaluation tools, setup guidance, a prerecorded synthetic demo, and this report.

Evidence note: evaluation metrics and successful live speech checks below are previously recorded candidate results. The packaging audit did not re-run dataset-enabled evaluations or make live provider calls.

System workflow

1. Input	2. Normalize	3. Ground	4. Respond
Audio or typed question	Speech/text normalization and language handling	Knowledge retrieval plus deterministic facts, calculations, and state	Answer or escalation; optional LLM wording and speech output

Main implementation components

Component	Role	Main files
Agent and conversation flow	Intent/rule handling, conversation context, escalation and tool workflow.	src/voiceops/agent.py, intent.py, tools_adapter.py
Retrieval and facts	Knowledge-base retrieval, product/order facts, calculations, text normalization, and scoring.	kb.py, facts.py, calc.py, textnorm.py, scorer.py
Speech and optional LLM	Gnani STT/TTS integration, optional LLM adapter, and safe bounded diagnostics.	voice.py, speech_text.py, llm.py, scripts/gnani_smoke_test.py
Interfaces and evaluation	Batch runner, text chat, browser demo, and offline evaluation tools.	run.py, chat.py, demo_server.py, eval/, tests/

Runtime note: the capstone dataset is supplied separately and is intentionally not inside this source package. Dataset-backed behavior and full evaluation require that dataset to be available and DATASET_DIR to point at its root.

Recorded evaluation results

The values in this section are transcribed from the supplied candidate documentation and report. They were not reproduced during this archive-only packaging audit.

Evaluation mode	Recorded outcome	Mean recall
Offline normal questions (52)	100.0% content accuracy; 100.0% decision accuracy	0.968
Offline ASR-text questions (52)	100.0% content accuracy; 100.0% decision accuracy	0.968
Offline conversations (96 turns / 23 conversations)	100.0% checks; 23/23 conversations fully correct	0.944
Bounded RAG normal questions (52)	98.1% content accuracy; 96.2% decision accuracy	0.968
Bounded RAG ASR-text questions (52)	100.0% content accuracy; 94.2% decision accuracy	0.968
Bounded RAG conversations (96 turns)	100.0% checks; 23/23 conversations fully correct	0.944

Bounded synthetic speech check

Operation	Previously recorded observation	Requests
Speech-to-text (STT)	Provider returned a nonempty transcript; transcript content was not printed or retained in the report.	1
Text-to-speech (TTS)	Generated a readable, nonempty WAV (2.24 seconds; 215,084 bytes).	1

The recorded STT clip was normalized to mono 16-bit PCM at 16 kHz; reported input duration was 12 seconds and RMS was 3912.6. The prior environment reported gnani-vachana 0.8.2. The live STT/TTS checks were not rerun for this packaging audit.

Evaluation interpretation and known limitations

• Recorded retrieval recall is below the saved candidate baseline: normal and ASR-text from 0.989 to 0.968, conversations from 1.000 to 0.944. Recorded rollback experiments did not restore the previous values.

• The latest recorded ASR-text RAG decision accuracy is 94.2%. A prior ASR RAG run had a provider transport error and used offline fallback; it is not counted as valid RAG evidence.

• The earlier STT failure was not reproduced, but its historical root cause is unknown. No fix to that failure is claimed.

• Browser microphone behavior and complete production end-to-end behavior were not verified. Embedding retrieval was not tested with a confirmed model configuration.

• The diagnostic held-out set was highly templated and is weak evidence of general retrieval quality. The reported metrics do not guarantee a competition score.

Package verification and reproducibility

This audit validates the submitted files and performs offline checks where possible. It does not substitute for running the project against the separately supplied dataset or performing live speech-provider checks.

Check	Result	Scope / interpretation
Python compilation	**PASS**	Compileall completed with exit code 0 across entry points, source, evaluations, scripts, and tests.
Offline pytest (dataset unavailable)	**26 passed; 43 skipped; 1 deselected**	Dataset-backed cases skipped because the dataset was absent. Hidden-case test needs the private eval/hidden_cases.csv.
Demo media	**PASS**	Readable MP4; H.264 video, AAC audio, 1280×720, about 27.54 seconds.
Technical report	**PASS**	Regenerated and rendered; visual inspection found no clipped text or overlap.
Manifest and ZIP	**PASS**	Final archive file list, sizes, SHA-256 hashes, and ZIP CRC checked.
Secret scan	**PASS, basic scan only**	No common key/token/private-key-header patterns found. This is not a comprehensive secrets audit.
Live provider calls	**NOT RUN**	No live Gnani or LLM requests were made during this packaging audit.

Reproduce offline checks (Windows PowerShell)

```
$env:DATASET_DIR = "C:\path\to\Capstone_Project_Dataset_FTTT_2026"
$env:VOICEOPS_NO_ENV_LOAD = "1"
$env:VOICEOPS_ENABLE_HYBRID_RAG = "0"
python -m compileall -q run.py chat.py demo_server.py src eval scripts tests
python -m pytest -q `
--deselect="tests/test_retrieval_v8.py::test_hidden_case_evidence_recall_at_seven_is_complete"
python "$env:DATASET_DIR\validate_dataset.py"
python eval/eval_questions.py --mode offline
python eval/eval_questions.py --mode offline --field asr_text
python eval/eval_conversations.py --mode offline
```

Submission contents and exclusions

The package contains application source, tests, evaluation scripts, setup instructions, report notes, demo video, technical report, and a manifest of file sizes and SHA-256 hashes. It excludes the separate dataset, private hidden cases/labels, real credentials, .env, .venv, Git history, Python bytecode, caches, logs, and generated evaluation-result JSON files.

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